

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Abstract

001 Long-horizon video generation is evaluated with whole-
002 frame metrics that reward motion and temporal consis-
003 tency. For fixed-camera nature scenes this creates an
004 ambiguity: motion of water, fire, smoke, or rain is de-
005 sirable, whereas motion of the background is an error.
006 A system can therefore score well on motion while its
007 scene drifts, or on consistency while its flow stagnates.
008 We introduce SNF-Bench, an evaluation framework
009 for long-horizon fixed-camera generation that parti-
010 tions each scene into static support and dynamic flow
011 and reports static fidelity, flow persistence with ab-
012 solute magnitude, and drift leakage separately, never
013 as one score. Each factor is validated mechanisti-
014 cally rather than by correlation with preference: we
015 inject global translation, rotation, and scale drift and
016 progressive motion attenuation at known severity into
017 real generations, and require each factor to respond
018 in its stated direction and to remain selective against
019 corruptions it does not target. Auditing publicly re-
020 leased long-horizon generators across text- and image-
021 conditioned tracks in their intended configurations, we
022 find that whole-frame motion and static-region drift in-
023 duce near-opposite orderings of the same outputs, and
024 that controlled drift which triples our static-fidelity fac-
025 tors moves a widely used whole-frame motion score by
026 roughly a tenth—in the direction that rewards it. SNF-
027 Bench measures where motion occurs and whether it
028 persists; it does not measure physical realism.

029 1. Introduction

030 Autoregressive and streaming video diffusion models
031 now extend generation from seconds to minute-scale
032 horizons. Evaluation has not kept pace: small tempo-
033 ral errors invisible in short clips accumulate over long
034 rollouts into background displacement, colour drift,
035 or motion decay, and whole-frame metrics summarise

these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature
scenes. In a static-camera video of a river, the wa-
ter should move while the bank, rocks, and surround-
ing landscape stay fixed. The same separation gov-
erns waterfalls, ocean waves, rainfall, fire, smoke, and
wind-driven vegetation: some regions form static sup-
port that must remain structurally fixed, while others
carry dynamic flow that must persist. Whole-frame
motion measures reward optical flow regardless of its
origin, crediting background drift as readily as genuine
fluid motion, while temporal-consistency measures can
favour videos whose stability comes from suppressing
the flow they should sustain. Two opposite failures—
static drift and flow decay—are thereby conflated, and
increasingly so at long horizons, where accumulated
background motion can dominate a sequence’s appar-
ent dynamics. Figure 1 shows both failures in released
generations of a single prompt.

We introduce SNF-Bench (Static–Natural Flow), an
evaluation framework built around this decomposition,
evaluated at 60 and 120s horizons with shorter and
longer tiers retained as diagnostics. Measurement is
organised into three factors. Static fidelity captures
geometric displacement and residual optical flow inside
nominally stationary regions. Flow persistence reports
the retention of dynamic-region motion together with
its absolute compensated magnitude, so a retention
ratio cannot appear strong merely because motion is
uniformly weak. Drift leakage relates static-region mo-
tion and global-motion compensation to the measured
dynamic-region flow.

A benchmark is only as trustworthy as its measure-
ments, so we validate the protocol independently of
any model comparison. Controlled perturbations in-
ject global translation, rotation, and scale drift, pro-
gressive motion attenuation, late freezing, photometric
drift, and temporal repetition into real fixed-camera
generations at known severity, and we require each fac-
tor to respond in the direction its definition states and

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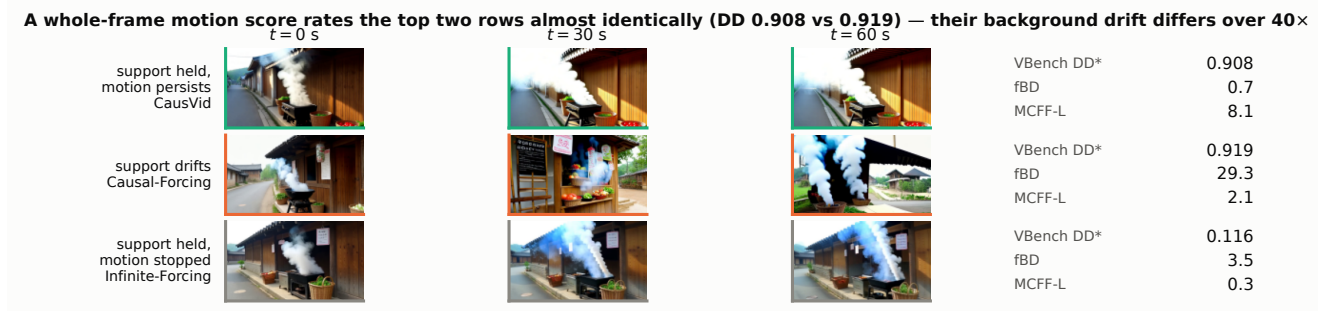


Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three released 60s generations of one prompt. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect. DD* is the per-video quantity underlying VBench Dynamic Degree, computed under its published rule but read before the binarisation the suite applies when aggregating, so that near-identical rows remain distinguishable as numbers.

to remain selective against corruptions it does not target. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit seven publicly released long-horizon generators in their intended configurations. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organised around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, and never as a single score.
3. Mechanistic validation of every reported factor under corruptions of known type and severity, with selectivity measured against the families each factor should ignore and every measurement recomputed under erosion and dilation of the region partition.
4. A public-model audit showing that this decomposi-

tion materially changes the interpretation and ordering of long-horizon behaviour relative to whole-frame metrics.

2. Related Work

Video-generation evaluation. General-purpose suites assess visual quality, temporal consistency, motion, and semantic alignment. VBench [8] decomposes quality into dimensions including background consistency, motion smoothness and dynamic degree, and VBench-2.0 [26] extends it toward intrinsic faithfulness; Eval-Crafter [13] learns a weighting of many metrics from user opinion; Video-Bench [4] scores generations with multimodal language models validated against human preference. Others specialise: T2V-CompBench [19] targets compositional binding, VMbench [11] perceptually aligned motion quality, and WorldScore [1] controllability and dynamics for world generation. These set the standards a benchmark paper must meet—explicit taxonomy, validated metrics, stated scale—but their measurements are whole-frame or sequence-level, and such an aggregate cannot reveal whether motion originates where intended or in displacement of nominally static structure. VMbench is the closest comparator, but it asks whether motion is perceptually plausible where we ask where it occurs and whether it survives compensation for global displacement. Our validation is correspondingly mechanistic—perturb by a known amount, require the expected response—complementing the human-alignment evidence those suites provide.

Long-horizon autoregressive generation. Autoregressive and streaming generators extend synthesis beyond short bidirectional clips by producing frames or tem-

poral chunks causally, including CausVid [24], Self-Forcing [7], Infinite-Forcing [9], Rolling-Forcing [12], Reward-Forcing [14], LongLive [22], and Causal-Forcing [27] with its few-step successor Causal-Forcing++ [25]; bidirectional systems such as LTX-Video [3] and Wan [21] provide short-horizon reference points. These systems differ substantially in temporal block size, training objective, memory and cache policy, and horizon-extension mechanism [23], yet their evaluation typically relies on generic video metrics or method-specific qualitative analysis. That diversity motivates auditing each system in its released configuration, and the evaluation gap motivates a system-agnostic question: does static support remain fixed while dynamic flow persists?

Image dynamics and nature-flow animation. Work on animating still images has long modelled natural motion explicitly: controllable fluid animation separates static image structure from specified fluid motion [15], and Generative Image Dynamics models long-term scene motion through learned trajectory representations [10]. These works treat static support and dynamic content as distinct modelling targets; SNF-Bench elevates that distinction to an evaluation principle for long-horizon generators.

3. SNF-Bench Design

SNF-Bench evaluates long-horizon generation under a deliberately constrained setting: a fixed camera observing a scene containing both stationary structure and persistent natural motion. This isolates two failure modes that are difficult to separate in unconstrained video—static drift, where nominally stationary structure moves over time, and flow decay, where intended dynamics weaken or disappear. Every sequence is treated as a composition of two spatial roles, Ω_{static} for regions expected to remain geometrically stationary and Ω_{flow} for regions expected to exhibit sustained motion. The partition defines an evaluation role, not a semantic ontology: a riverbank is static support while the river surface is flow; buildings are static support in a rainfall scene while the falling rain is flow. SNF-Bench does not require dynamics inside Ω_{flow} to follow any physical model. It measures whether motion stays localised to the appropriate regions and persists over time without contaminating static support.

Every system receives identical conditioning per item, and no prompt is rewritten for a particular method. Both conditioning modalities are evaluated under the same decomposition, differing only in how static support is established: the text-conditioned setting builds it in the opening of the sequence, while the

image-conditioned setting takes it from a shared source image, so divergence there is attributable to the system rather than to a differently imagined scene. Both tracks are reported in the main text at the 60 and 120 s horizons; all four horizons appear in the supplementary material.

3.1. Scene Categories and Horizons

Prompts are stratified across representative fixed-camera natural processes—channel water, ocean waves, precipitation, fire and smoke, lava, and wind-borne motion—with the category set fixed before the final sweep. The strata are not equally populated: channel water accounts for 61% of the principal prompt set. Reported aggregates are means over prompts, not category macro-averages: three of the five populated categories hold two prompts each, so equal category weighting would buy balance at a cost in variance the sample cannot support. The supplementary material instead recomputes every system both ways and states which conclusions survive, with complete per-category counts.

Four temporal regimes are used: 5 s as an initialisation check, 60 s as the primary reported horizon, 120 s as a supporting tier, and 240 s as a diagnostic extreme. Conclusions rest on the 60 s tier, which carries 23 prompts; the 120 s tier is reported alongside it as a supporting check, and the 5 and 240 s tiers—an initialisation check and a diagnostic extreme—are in the supplementary material. A system that cannot reach a horizon under its released configuration is reported as unsupported rather than silently modified.

3.2. Region Partition

The partition is obtained automatically and identically for every sequence. Mean optical-flow magnitude over the leading 12% of sampled frame pairs is thresholded by Otsu’s method [16] into a dynamic region and its complement; the static region is then eroded and a 4% frame border discarded, which together form an ignored transition band so that boundary pixels—where the partition is least certain and flow least reliable—contribute to neither region.

Restricting the estimate to an early window prevents circularity: deriving a partition from the entire motion field would let accumulated drift be read as evidence that the affected region is dynamic. At 60 s the partition is fixed from the first 7.1 s and the metrics evaluated over the remainder, so it reflects each system’s layout—unavoidable when every system authors its own scene—but not the temporal failure being measured.

Two limitations bound the later conclusions. The

partition is automatic, so a system drifting within the first 12% of its rollout can influence its own partition; we therefore treat partitions as measured quantities and recompute every result under systematic erosion and dilation of the boundary (Sec. 5). And a two-way partition cannot represent layered content: rain and snow transit static support, so those pixels are assigned by flow magnitude alone and static-fidelity values in precipitation scenes carry an unmodelled contribution. Derivation, the sensitivity study, and the annotation protocol we regard as the natural strengthening of this component are in the supplementary material.

259 3.3. Benchmark Manifest

SNF-Bench is defined by a versioned manifest fixed before the final sweep, recording per item: identifier and category, exact prompt, target horizons, resolution and frame-rate metadata, and provenance. After freezing, items are never removed because a system performs poorly; corrections require a new benchmark version. A failure taxonomy—unsupported horizon, out-of-memory, runtime crash, numerical corruption, invalid output—is likewise fixed in advance, and failures are reported alongside successful generations rather than silently excluded. Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial.

274 4. Drift–Motion Evaluation Protocol

SNF-Bench evaluates fixed-camera generation along three complementary factors: Static Fidelity, Flow Persistence, and Drift Leakage. No individual score is sufficient: low background motion can be achieved by freezing the entire video, and high foreground motion can be produced by drifting the entire scene. We therefore always report static-region stability jointly with both the relative persistence and absolute magnitude of dynamic-region motion.

Let I_t denote frame t , Ω_{static} and Ω_{flow} the region masks of Sec. 3.2, and $\mathbf{u}_t$ the optical flow from I_t to I_{t+1} , estimated with RAFT [20]. Each sequence defines an early window $\mathcal{E}$ and a late window $\mathcal{L}$ of equal duration, $\mathcal{E}$ beginning after a warm-up so that generation transients do not define the motion reference; extents are fixed in the specification.

291 4.1. Static Fidelity

Feature-aligned Background Drift (fBD). ORB features [18] extracted inside Ω_{static} from the first frame are matched against each late-window frame under robust geometric filtering; fBD ($\downarrow$) is the median matched-point displacement, averaged over the late

window and expressed as a percentage of the frame diagonal. It captures accumulated geometric displacement of persistent structure and is intentionally insensitive to fine-scale flicker that moves no structure (formal definition in Supp. ??).

Normalized Background Flow (NBF). Complementing accumulated drift, NBF ($\downarrow$) is the mean optical-flow magnitude inside the static mask, averaged over all frame pairs and normalised by frame width for resolution invariance and by the inter-frame interval Δt for frame-rate invariance, giving units of 10^{-3} frame widths per second (Supp. ??). Here Δt is the interval of the sampled pairs actually used for flow estimation: methods are evaluated at their native frame rates, which differ across the roster, and the evaluation pipeline subsamples each clip to a common temporal rate before measurement. Expressing NBF per unit time rather than per frame makes the quantity independent of both the native rate and the sampling policy, so the measure remains comparable if either changes.

NBF is a normalised magnitude, not a ratio: dividing by whole-frame flow would couple background leakage to foreground behaviour and inflate the measure for methods whose dynamic motion decays.

512 4.2. Flow Persistence

Global-motion compensation. Coherent scene displacement is removed before dynamic-region motion is evaluated. A single translation vector is insufficient: under rotation or zoom the static-region flow field is not constant, and its component-wise median is near zero for a centered rotation, so a translation-only estimate would leave exactly the corruptions Sec. 5 injects uncompensated. We therefore estimate a robust global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ from static-region correspondences under RANSAC, and subtract the displacement field it induces from the measured flow pointwise, giving a compensated field $\tilde{\mathbf{u}}_t$ (Supp. ??). A similarity rather than an unrestricted affine model is used because affine shear absorbs legitimate local deformation and over-corrects. Where correspondences are insufficient we fall back to median translation and record the fallback per sequence.

Motion-Compensated Foreground Flow (MCFF). MCFF ($\uparrow$) is the mean compensated flow magnitude inside Ω_{flow} over a temporal window. We report it on both the early and the late window—MCFF-E and MCFF-L—so a method with uniformly negligible motion cannot appear strong through relative measures alone (Supp. ??).

Flow Persistence (FP). FP ($\uparrow$) is the ratio of late to early MCFF, clipped at a fixed constant to prevent unstable ratios when early motion is small (Supp. ??). FP near one indicates retained motion; substantially smaller values indicate decay; values above one are not automatically better. FP is never reported without its MCFF magnitudes.

4.3. Drift Leakage and Attenuation

Whole-frame metrics cannot determine whether apparent dynamic-region motion arises from intended flow or from displacement of the whole scene; we report two diagnostics, since neither subsumes the other. Let F_{static} denote mean flow magnitude over Ω_{static} , and F_{raw} , F_{comp} the mean raw and compensated magnitude over Ω_{flow} , all in the late window, where drift has accumulated.

Drift Leakage Ratio (DLR). DLR ($\downarrow$), the ratio of static-region to raw dynamic-region flow magnitude in the late window, locates the sequence’s motion energy (Supp. ??). It is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates that static support moves more than the region intended to move, a regime that occurs in practice.

Drift Attenuation Ratio (DAR). DAR ($\downarrow$), one minus the ratio of compensated to raw dynamic-region flow, measures the signed relative change in dynamic-region motion after compensation—how much apparent motion a global displacement model accounts for (Supp. ??). Reported values are clipped to $[0, 1]$ while the signed quantity is retained for validation; negative values, which arise where local flow opposes the estimated global field, are reported rather than suppressed.

Interpretation. Neither is a causal decomposition of motion into drift and non-drift components, and we do not describe them as one. Read together: high DLR with moderate DAR indicates static-region motion energy a rigid global model does not explain, implicating non-rigid instability such as warping rather than uniform translation.

Semantic guardrail. The decomposition is agnostic to what moves: a sequence rendering drifting fog where a river was requested can occupy a favourable operating point. The natural guardrail is a prompt-alignment score—CLIP-based image-text similarity [6], or a learned video-quality predictor [5] at higher cost. We do not report one, and the factors should be read accordingly: they establish where motion occurs and whether it persists, not that the depicted phenomenon

is the one requested. Pairing SNF-Bench with a semantic score is how it should be used in practice.

4.4. Joint Interpretation

No factor is read alone: low drift with low compensated motion indicates an over-stabilised sequence, high compensated motion with high static-region flow indicates motion contaminated by displacement, and high retention with negligible magnitude indicates persistent but trivial motion. SNF-Bench therefore reports no single scalar score, since collapsing the factors would recreate the ambiguity it exists to expose.

Implementation convention. Videos are normalised to a common evaluation resolution and subsampled to a common flow-estimation rate; no frame interpolation is performed, since synthesising frames would alter the flow field being measured. Every estimator, threshold, window extent and constant is fixed globally before the sweep and applied unchanged to every system, and the per-sequence frame rate, resolution and compensation mode are released alongside the scores.

5. Controlled Validation

Before SNF-Bench is used to compare generators, we validate that its measurements respond to the failure modes they target. A benchmark is not justified merely because its metrics rank differently; each must change selectively, monotonically, and in the expected direction under known corruptions. We therefore construct controlled variants of fixed-camera reference clips with known corruption type and severity, independently of the audit in Sec. 6: perturbation strengths, masks, windows, and hyperparameters are fixed before the final results are inspected.

5.1. Perturbations and Criteria

Given a reference sequence we generate variants that isolate individual components of the decomposition. Geometric drift applies a temporally increasing global transformation—translation, rotation, or scale—simulating camera-like background displacement that accumulates as long-horizon error does. Motion attenuation progressively blends dynamic-region content toward an early reference, and late freezing replaces all content after a chosen fraction of the sequence, both leaving static support untouched. Photometric drift varies brightness and colour temperature while preserving geometry and motion, bounding the residual sensitivity of flow-based quantities. Temporal repetition replaces late dynamic content with a short repeated cycle. Partition perturbation erodes and dilates the region boundary rather than the video.

Geometric severity is specified as the mean pixel displacement a corruption induces, with each family's parameter derived from it using the frame geometry. This is not presentational: a parameter in degrees is not commensurate with one in pixels, and at the evaluation resolution one degree of rotation induces only a few pixels of displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude.

A factor is admitted only if its response direction matches its definition, that response is monotonic in severity, unrelated corruptions produce substantially smaller changes, and conclusions survive perturbation of the region partition. Factors failing these checks are reported as diagnostics rather than headline quantities. The severity specification and the admission outcome for every factor are in the supplementary material.

Complementary checks. Mechanistic validation shows a factor responds to the corruption it names, not that its ordering matches what an observer perceives, and we do not close that gap. The supplementary material specifies an axis-specific protocol—forced choice on one failure axis, never on which output is better—and reports the one instance we ran: an open-weight judge [17] failed a presentation-order control, so no agreement rate is drawn from it. Perceptual agreement remains open.

5.2. Validation Results

Figure 2 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive attenuation of dynamic-region content drives MCFF-L and FP down while leaving the static factors comparatively unchanged—the selectivity the decomposition requires. The comparison that motivates the benchmark appears in the same experiment. Injected displacement that carries fBD to $3.2\times$ its unperturbed value moves VBench Dynamic Degree to only $1.1\times$, and in the wrong direction: a whole-frame motion score is largely insensitive to background drift, and to the extent it responds at all it improves on precisely the corruption a fixed-camera evaluation should penalise.

Two results qualify the framework rather than support it. Rotation is the harder geometric case for every factor, since correspondences degrade and the induced displacement is smallest at the frame centre: both static-fidelity factors respond more weakly than under translation ($+0.70$ for fBD, $+0.60$ for NBF, against $+1.00$ for both), so neither is used alone. DLR is the sharper limitation—admitted on its target family, where it responds perfectly, but with an inverted ro-

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	✓

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Twelve reference clips per severity level.

tational response (-0.80), so it is read only as a comparative diagnostic alongside DAR. Reported alongside these is the scope boundary of Sec. 5.1: looping content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine progression from repetition, and does not claim to.

6. Public-Model Audit

We apply SNF-Bench to publicly released long-horizon generators. The goal is not a universal leaderboard but a determination of how deployed systems occupy the static-fidelity/flow-persistence space when used as their authors intended.

Systems and setting. The audit covers 7 publicly released autoregressive or streaming generators with accessible checkpoints and reproducible inference code, frozen as a set before the sweep. Unpublished and internal systems are excluded by construction, as are systems whose outputs cannot be independently reproduced. Each is run in the deployment regime its release intends: differences in temporal block size, denoising schedule and memory policy are preserved deliberately, being part of the released system rather than nuisance variables. The audited systems share a deployment envelope—all generate 832×480 at 16 fps and reach 240s without modification to their released temporal policy—so no comparison here is confounded by output format. We never substitute a block size, cache policy or schedule to populate a table entry; a system that cannot reach a horizon unmodified is marked unsupported.

The image-conditioned track is audited under the

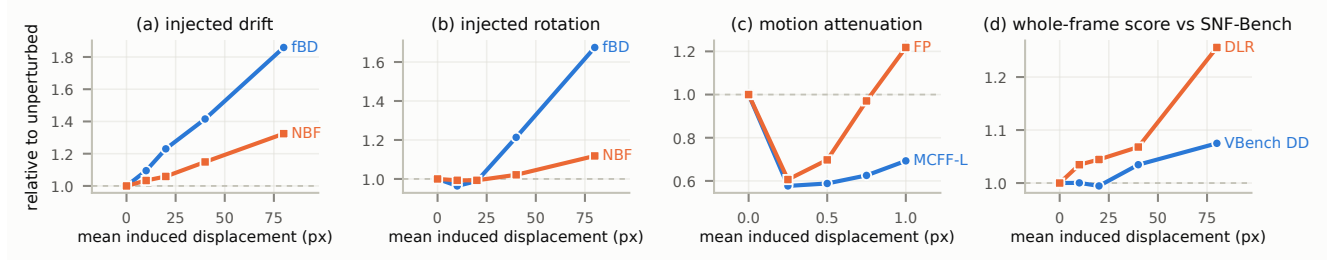


Figure 2. Mechanistic validation. Each panel injects a corruption of known kind and severity into real fixed-camera clips and plots each factor relative to its unperturbed value. Panel (d) puts VBench Dynamic Degree through the identical corruption as the SNF factors. Displacement that raises fBD to $3.2\times$ its unperturbed value moves the whole-frame score by $1.1\times$ —and upward: the generic score is close to blind to where the motion originates, and what little it registers it counts in the drifting system’s favour. The perturbation is synthetic, so the comparison implicates no published method.

526 same protocol and the same frozen specification. Sys-
527 tems evaluated under a common long-horizon wrapper
528 rather than their released configuration are labelled
529 matched in every table, and their scores are read as
530 deployment sensitivity rather than as the published
531 method’s performance; we never rank a native row
532 against a matched one.

533 Controls and statistics. Within each item every sys-
534 tem receives identical conditioning, and the seed set is
535 fixed before the sweep. Seed control guarantees repro-
536 ducibility within a system, but because implementa-
537 tions consume randomness differently, identical seeds
538 do not induce matched noise draws across systems;
539 cross-system comparison therefore rests on the paired-
540 item design. Videos are retained in native form and
541 the normalisation of Sec. 4 applied unchanged.

542 Scores are computed per item and averaged over
543 items; the sensitivity of every conclusion to category
544 macro-averaging is reported in the supplementary ma-
545 terial. Uncertainty is estimated by percentile boot-
546 strap [2] over items, and we report 95% intervals. Ex-
547 act ordinal ranks are avoided where uncertainty does
548 not support them: systems whose intervals overlap
549 substantially are not described as separated. Op-
550 erational limits—unsupported horizons, failed gener-
551 ations, memory exhaustion, invalid outputs—are re-
552 ported as rates against the frozen manifest and never
553 removed after inspection.

554 To test whether the decomposition changes interpre-
555 tation, the same outputs are additionally scored with
556 established whole-frame metrics and the induced or-
557 derings compared. The focus is interpretation change,
558 not a claim that existing metrics are wrong.

7. Results and Interpretation Changes

Unless stated otherwise, comparisons refer to systems
in their released configurations; deployment-sensitivity
results are reported separately in the supplementary
material and never redefine a system’s native be-
haviour.

7.1. Native-Capability Audit

Table 2 reports the text-conditioned track at 60s, the
tier whose sample supports method-level separation,
and at 120s as a supporting tier. The 120s panel car-
ries 6 prompts and is read as corroboration of direction
rather than as an independent separation; it draws a
different prompt set from the 60s panel, so the two
panels are not compared column-by-column. The 5
and 240s tiers, an initialisation check and a diagnostic
extreme respectively, are in the supplementary mate-
rial with all four horizons.

Motion persistence. Dynamic-region behaviour is
read at the joint operating point (MCFF(L), FP),
never FP alone: sustained flow requires non-trivial late
motion and high retention. Reporting MCFF-E beside
MCFF-L exposes the degenerate case, since a system
with negligible motion throughout cannot reach a high
FP without a small MCFF-E.

Operating regimes. Read jointly, the static-fidelity
and persistence columns separate low-drift/low-
motion, high-motion/high-drift and low-drift/non-
trivial-motion regimes without collapsing the axes into
a scalar score; the supplementary material plots them.
The extremes are informative in opposite directions.
Infinite-Forcing has the lowest normalised background
flow, and is the only system whose interval on that fac-
tor excludes every other, while also showing the least
surviving late motion; on feature-based drift its interval

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 23$)				
CausVid	8.08 ± 3.15	9.27 ± 4.14	2.87 → 1.96	0.67 ± 0.18
Self-Forcing	11.4 ± 3.63	12.7 ± 3.97	10.4 → 2.54	0.44 ± 0.22
Infinite-Forcing	7.75 ± 3.84	3.65 ± 0.88	1.17 → 0.29	0.63 ± 0.22
Rolling-Forcing	16.8 ± 3.69	11.2 ± 2.70	3.18 → 1.57	0.75 ± 0.29
Reward-Forcing	14.8 ± 3.71	8.51 ± 2.79	3.81 → 0.80	0.49 ± 0.18
LongLive	15.9 ± 2.89	13.9 ± 4.76	6.46 → 2.41	0.75 ± 0.28
Causal-Forcing	22.4 ± 2.26	87.6 ± 18.6	9.84 → 7.21	0.83 ± 0.31
120s horizon ($n = 6$)				
CausVid	6.15	3.42	0.36 → 0.52	1.06
Self-Forcing	21.2	6.80	4.85 → 0.82	0.85
Infinite-Forcing	16.1	2.60	0.50 → 0.32	0.64
Rolling-Forcing	23.8	10.6	3.33 → 0.77	0.38
Reward-Forcing	20.0	11.6	2.05 → 0.69	0.39
LongLive	22.5	9.66	5.10 → 1.19	0.81
Causal-Forcing	21.8	130	10.5 → 10.7	0.90

Table 2. SNF-Bench audit, T2V track, native configuration. All systems are public external models run under their own intended configuration. Our own systems are excluded by construction. DAR is reported clipped to $[0, 1]$; signed values are released. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel. Values are means over prompts $\pm$ half a percentile bootstrap 95% interval (10k resamples, fixed seed); panels with fewer than ten prompts report the point estimate alone, since an interval resampled from so few items is not informative. MCFF-E and MCFF-L are given together because FP is a ratio and is not interpretable without the magnitude it is taken over; intervals overlap for several systems, and only non-overlapping comparisons are described as separated. Aggregation sensitivity and the DLR/DAR diagnostics are reported in the supplementary material.

593 overlaps the next-ranked system, so it is not the least-
594 drifting system outright. Causal-Forcing sits alone at
595 the opposite corner, separated on background flow by
596 a wide margin.

597 7.2. Drift Leakage Changes the Interpretation of 598 Apparent Motion

599 The effect is systematic rather than anecdotal. Corre-
600 lating each generic whole-frame metric with each SNF-
601 Bench factor over method-level means (supplementary
602 material), Dynamic Degree tracks static-region flow at
603 $\rho = +0.93$, so the systems a generic metric reads as
604 most dynamic are, closely, those whose backgrounds
605 move most. Causal-Forcing ranks 1 of 7 on Dynamic
606 Degree and 7 on NBF; Infinite-Forcing mirrors it, rank-
607 ing 1 on static fidelity and 7 on apparent motion. Read
608 with its magnitudes, Causal-Forcing’s DLR of 0.94 in-
609 dicates static-region motion energy almost equal to the
610 region intended to move, while its DAR of 0.40 indi-
611 cates a rigid global model accounts for only part of
612 it—the remainder is non-rigid instability.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 30$)				
Causal-Forcing++	9.18	5.97	— → 0.67	0.86
(2-step) [†]				
Causal-Forcing++	5.34	7.05	0.86 → 0.57	0.77
(1-step) [†]				
Causal-Forcing	21.1	36.4	2.80 → 2.69	0.89
(frame-wise) [†]				
Causal-Forcing++	9.94	6.91	— → 0.81	0.98
(2-step, frame-wise)				
Self-Forcing [†]	17.3	5.52	— → 0.74	0.88
CausVid [†]	19.9	3.31	0.58 → 0.41	0.81
Causal-Forcing++	10.4	5.25	— → 0.52	0.97
(1-step, frame-wise)				
120s horizon ($n = 20$)				
Causal-Forcing++	9.65	8.81	— → 0.93	0.91
(2-step) [†]				
Causal-Forcing++	7.26	9.39	— → 0.83	0.84
(1-step) [†]				
Causal-Forcing	19.9	52.7	4.82 → 5.06	1.08
(frame-wise) [†]				
Causal-Forcing++	11.4	17.3	— → 1.54	0.96
(2-step, frame-wise)				
Self-Forcing [†]	23.0	10.3	— → 0.78	0.44
CausVid [†]	19.7	6.04	1.15 → 0.83	0.67
Causal-Forcing++	13.6	7.39	— → 0.43	0.68
(1-step, frame-wise)				

Table 3. SNF-Bench audit, I2V track. Setting is stated per row: native systems run under their authors’ configuration, matched systems under the common long-horizon wrapper and are read as a deployment stress test rather than as the published method’s performance. [†] marks a system run under the common long-horizon wrapper rather than its released configuration; such rows are read as deployment sensitivity and are never ranked against native rows. n is stated per horizon and is common to every row of that panel. Values are means over prompts; intervals for this track are given with the complete per-horizon tables in the supplementary material. MCFF-E and MCFF-L are given together because FP is a ratio and is not interpretable without the magnitude it is taken over; intervals overlap for several systems, and only non-overlapping comparisons are described as separated. Aggregation sensitivity and the DLR/DAR diagnostics are reported in the supplementary material.

613 7.3. Image-Conditioned Track

614 Table 3 reports the image-conditioned track at the
615 same two horizons. Unmarked rows run under their
616 authors’ released configuration and are comparable to
617 the text-conditioned audit; marked rows run under a
618 common long-horizon wrapper, because the released
619 configuration does not reach these horizons, and are
620 read as deployment sensitivity rather than as the pub-
621 lished method’s performance. Ranking across the two
622 would compare one system’s design against another’s
623 deployment, so we do not.

624 The track supplies what the text-conditioned one
625 cannot: every system starts from the same source im-
626 age, so static support is given rather than invented and

divergence is attributable to the system. Bidirectional reference systems reach only the 5s tier and are reported there.

8. Limitations and Conclusion

SNF-Bench is intentionally narrow: long-horizon fixed-camera generation, where static support and dynamic flow separate. It measures neither physical realism nor semantic correctness, cannot distinguish progression from repetition, and rests on an automatic partition whose limits are stated in Sec. 3.2 and bounded by the erosion-dilation study. Conclusions unstable under that perturbation are never headline claims.

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